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5

INELASTIC COLLISIONS AND SURFACE INTERACTIONS

5.1 Molecules with rotational energy

As noted in Chapter 1, rotational energy is associated only with diatomic and polyatomic molecules that consist of two or more atoms. Furthermore, there is no rotation about axes with a small moment of inertia, such as the internuclear axis of diatomic molecules. The most realistic of the classical monatomic models is the point centre of force, and the obvious extension would be two or more point centres of force separated by fixed distances.

Lordi and Mates (1970) studied the *two centres of repulsion* model and found that a complex numerical solution was required for each set of impact parameters. The model does not yield closed form expressions for the transport properties and it is impractical for application to simulation studies that require the computation of many millions of typical collisions. The *sphero-cylinder* model consists of a smooth elastic cylinder with hemispherical ends and is a logical extension of the simple hard sphere model with apparent relevance to diatomic gases. The collision mechanics of sphero-cylinders has been developed by Curtiss and Muckenfuss (1958). The model requires additional variables for the specification of the spatial orientation as well as the angular velocity. A particular difficulty is that this orientation, and therefore the cross-section, changes with time. Whether or not two molecules collide depends not only on the miss distance, but also whether or not they 'mesh' or 'clash'. Moreover, there may be multiple 'chattering' impacts in a single collision event. The *weighted-sphere* model had already been suggested by Jeans (1904) and has been extensively developed by Dahler and Sather (1962) and Sandler and Dahler (1967). Although it is spherical in geometry, the molecule rotates about the offset centre of gravity rather than the geometric centre. It therefore suffers from essentially the same disadvantages as the sphero-cylinder model. The only exact physical model of a rotating molecule that has a cross-section that is not affected by its orientation is the *rough-sphere* model. The name of this model is based on its basic physical property which is that the velocities at the point of contact of the two molecules are reversed on impact. The model was first suggested by Bryan (1894) and the transport properties were analysed by Pidduck (1922). This is the most easily applied model and it is described in detail in §5.2. The transport properties have been presented and critically discussed by Chapman and

Cowling (1970). As with the hard sphere model for a monatomic gas, the coefficient of viscosity is proportional to the square root of the temperature. It is more unrealistic than the hard sphere model in that a glancing collision can result in a large deflection. Also, all collisions can result in a large interchange of rotational and translational energy and the rotational relaxation time is unrealistically short. This could be lengthened by specifying that some fraction of the collisions follow the hard sphere, rather than rough-sphere, mechanics. However, there does not seem to be any way around the main disadvantage of the model. This is that it possesses three internal degrees of freedom, while most applications call for the two degrees of freedom of the common diatomic gases. The limitations of the models based on exact dynamical systems are sufficient to justify recourse to a phenomenological approach.

One way of dealing with internal states is to set cross-sections for all the allowable transitions between the rotational energy levels. The number of levels is of the order of the ratio of the temperature to the characteristic temperature of rotation Θ_r, that is tabulated in Appendix A. Except in the case of hydrogen or deuterium, the number of rotational levels that must be considered is sufficiently large to make the approach based on the cross-sections for the individual transitions excessively cumbersome. It also means that the rotational modes are fully excited, and that a classical representation is valid. The phenomenological models superimpose the energy exchange between the internal and translational energy upon an otherwise monatomic model. The models are therefore spherically symmetric, but since real collisions tend to average over all molecular orientations, and there can be no preferred directions in an equilibrium gas, this should not be a serious limitation. The phenomenological processes must allow the specification of the requisite number of rotational degrees of freedom and lead to equipartition between these modes. There must also be exact equipartition between the rotational and translational modes and, when the gas is in equilibrium, the distribution function of each quantity must be the local Maxwellian. In practice, this means that the processes must satisfy the principle of reciprocity or detailed balancing. It must also allow the specification of the rotational relaxation time, including the temperature dependence of this time.

The most widely used phenomenological model is that of Borgnakke and Larsen (1975), and it is described in detail in §5.3. Detailed balance is ensured by the selection of the post-collision values directly from the Maxwellian distributions. The relaxation rate is controlled by regarding a fraction of the collisions as elastic. This method is quite unrealistic from the physical point of view, but satisfies all requirements for a successful phenomenological model. The 'energy sink' model (Bird 1976) is more realistic from the physical point of view and forces equipartition. However, it cannot be recommended because it leads to distortions of the equilibrium distribution function and does not satisfy detailed balancing. Pullin (1978) reviewed the phenomenological models and introduced a more realistic, but more cumbersome, development of the Larsen–Borgnakke approach.

5.2 The rough-sphere molecular model

As noted above, the essential feature of this model is that, in a collision, the relative velocity at the point of contact is reversed. Both the translational and rotational velocities contribute to this relative velocity and its reversal generally leads to an interchange of energy between the rotational and translational modes, as well as between the collision partners. The total energy is, of course, conserved.

The rough sphere is the simplest of the real physical models with rotation and is useful for comparative tests with phenomenological models. The phenomenological models may be set to have identical macroscopic properties, and the purpose of the tests is to determine whether there is then any difference in the results for typical flows. A simple gas generally suffices for such tests, and the analysis assumes that the molecules are identical. The objective is to determine the relationship between the pre-collision and post-collision values of the translational and rotational velocity components, as a function of the impact parameters.

The angular velocity of a molecule is denoted by $\boldsymbol{\omega}$ and its components about the x-, y-, and z-axes are ω_x, ω_y, and ω_z, respectively. The rotational energy of a molecule is $\frac{1}{2}I\omega^2$, where I is the moment of inertia. The moment of inertia depends upon the radial mass distribution of the molecule and therefore allows a limited range of variability in the model. It is most convenient to take a uniform mass distribution, for which

$$I = md^2/10. \tag{5.1}$$

The combined distribution function is (Chapman and Cowling 1970)

$$f = \frac{(mI)^{3/2}}{(2\pi kT)^3} \exp\left(-\frac{mc'^2 + I\omega^2}{2kT}\right)$$

and this may be written

$$f = (\beta^3/\pi^{3/2}) \exp(-\beta^2 c'^2)\{1/(\pi^{3/2}\omega_m^{3/2})\} \exp(-\omega^2/\omega_m^2). \tag{5.2}$$

Here, ω_m is the most probable angular speed and is related to the temperature by

$$\omega_m = (2kT/I)^{1/2}, \tag{5.3}$$

or, using eqn (5.1) for the uniform mass distribution,

$$\omega_m = (20RT)^{1/2}/d. \tag{5.4}$$

The equilibrium distribution of the angular speed ω with respect to the most probable value is, therefore, completely analogous to the distribution of the thermal speed with respect to the most probable molecular speed. All directions are equally probable for the axis of rotation. A rotational temperature may be defined by

$$\tfrac{3}{2}kT_{rot} = \tfrac{1}{2}I\overline{\omega^2}. \tag{5.5}$$

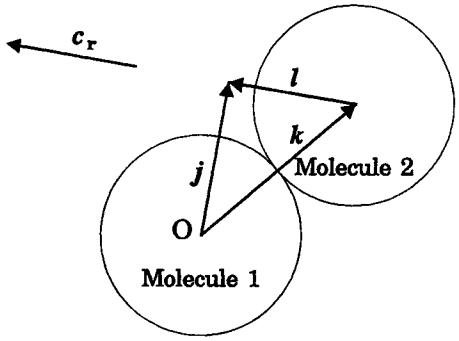


Fig. 5.1 The vectors $\boldsymbol{j}$, $\boldsymbol{k}$, and $\boldsymbol{l}$ in a rough-sphere collision.

The pre-collision values of the translational and rotational velocities of a collision pair are $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, and $\boldsymbol{\omega}_1$, $\boldsymbol{\omega}_2$, respectively. These are regarded as known quantities and the objective is to determine the post-collision values $\boldsymbol{c}_1^*$, $\boldsymbol{c}_2^*$, $\boldsymbol{\omega}_1^*$, and $\boldsymbol{\omega}_2^*$.

The pre-collision relative velocity vector $\boldsymbol{c}_r$ is defined here by

$$\boldsymbol{c}_r = \boldsymbol{c}_2 - \boldsymbol{c}_1.$$

Note that the sign convention is the opposite to that in eqn (2.3). The miss-distance impact parameter b is distributed between 0 and d, with probability proportional to d, while the azimuthal parameter ε is uniformly distributed between 0 and 2π. It is convenient to define three vectors $\boldsymbol{j}$, $\boldsymbol{k}$, and $\boldsymbol{l}$ in the collision plane, as shown in Fig. 5.1. The vector $\boldsymbol{k}$ is defined by the line joining the centre of molecule 1, which is taken as the origin O, to the centre of molecule 2. The vector $\boldsymbol{l}$ is in the direction of the relative velocity vector $\boldsymbol{c}_r$ and joins the centre of molecule 2 to the plane through O that is normal to the direction of $\boldsymbol{c}_r$. Finally, the impact vector $\boldsymbol{j}$ is defined by $\boldsymbol{j} = \boldsymbol{k} + \boldsymbol{l}$ and its magnitude is equal to the impact parameter b. Therefore, since the magnitude of $\boldsymbol{k}$ is equal to d and $\boldsymbol{l}$ is normal to $\boldsymbol{j}$,

$$l = (d^2 - b^2)^{1/2}. \tag{5.6}$$

Angles θ and ϕ are now defined such that θ is the angle between the x-axis and $\boldsymbol{c}_r$, and ϕ is the angle between the y-axis and the plane containing the x-axis and $\boldsymbol{c}_r$. Therefore

$$\cos\theta = u_r / c_r$$

and

$$\cos\phi = v_r / (c_r \sin\theta). \tag{5.7}$$

Then, with the azimuthal impact parameter measured from the plane $y = 0$, the components of the impact vector are

$$j_x = b\cos\varepsilon\sin\theta,$$

$$j_y = -b(\cos\varepsilon\cos\theta\cos\phi + \sin\varepsilon\sin\phi),$$

and (5.8)

$$j_z = b(\sin\varepsilon\cos\phi - \cos\varepsilon\cos\theta\sin\phi).$$

The magnitude of $\boldsymbol{l}$ is given by eqn (5.6) and its components are

$$l_x = l u_r / c_r,$$

$$l_y = l v_r / c_r,$$

and (5.9)

$$l_z = l w_r / c_r.$$

The components k_x, k_y, and k_z of the vector $\boldsymbol{k}$ along the line of centres follow from the relation $\boldsymbol{k} = \boldsymbol{j} - \boldsymbol{l}$. The components of the relative velocity of the point of contact are then given by

$$g_x = u_r + \tfrac{1}{2}\{k_y(\omega_{z,1} + \omega_{z,2}) - k_z(\omega_{y,1} + \omega_{y,2})\},$$

$$g_y = v_r + \tfrac{1}{2}\{k_z(\omega_{x,1} + \omega_{x,2}) - k_x(\omega_{z,1} + \omega_{z,2})\},$$

and (5.10)

$$g_z = w_r + \tfrac{1}{2}\{k_x(\omega_{y,1} + \omega_{y,2}) - k_y(\omega_{x,1} + \omega_{x,2})\}.$$

The scalar product of $\boldsymbol{k}$ and $\boldsymbol{c}_r$ is denoted by k_c, so that

$$k_c = \boldsymbol{k}.\boldsymbol{c}_r = k_x u_r + k_y v_r + k_z w_r.$$

The post-collision values of the translational velocity components may now be related to the pre-collision values by

$$u_1^* - u_1 = u_2 - u_2^* = \tfrac{2}{7}g_x + \tfrac{5}{7}k_c k_x / d^2,$$

$$v_1^* - v_1 = v_2 - v_2^* = \tfrac{2}{7}g_y + \tfrac{5}{7}k_c k_y / d^2,$$

and (5.11)

$$w_1^* - w_1 = w_2 - w_2^* = \tfrac{2}{7}g_z + \tfrac{5}{7}k_c k_z / d^2,$$

Finally, the post-collision angular velocity components are given by

$$\omega_{x,1}^* - \omega_{x,1} = \omega_{x,2}^* - \omega_{x,2} = \tfrac{10}{7}(k_y g_z - k_z g_y)/d^2,$$

$$\omega_{y,1}^* - \omega_{y,1} = \omega_{y,2}^* - \omega_{y,2} = \tfrac{10}{7}(k_z g_x - k_x g_z)/d^2,$$

and (5.12)

$$\omega_{z,1}^* - \omega_{z,1} = \omega_{z,2}^* - \omega_{z,2} = \tfrac{10}{7}(k_x g_y - k_y g_x)/d^2.$$

5.3 The Larsen–Borgnakke model in a simple gas

If a collision is regarded as inelastic, the total energy is reassigned between the translational and internal modes by sampling from the equilibrium distributions of these modes that are appropriate to this total energy. The relaxation rate is controlled through the fraction of collisions that are regarded as inelastic. In fact, the inverse of this fraction provides a first approximation to the relaxation collision number. The elastic collisions are calculated as monatomic collisions. When this model is applied in simulation methods that require the computation of millions of collisions, it has near optimal computational efficiency.

In order to improve the physical realism of the model, Larsen and Borgnakke (1974) introduced the 'restricted energy exchange' modification. In this, all collisions are treated as inelastic, but the proportion of the energy of the collision pair that participates in the energy exchange is restricted. This is less efficient from the computational point of view and, more seriously, Pullin (1978) showed that it does not satisfy detailed balancing. We will be concerned only with the procedures for the completely inelastic collisions that are applied to some fraction Λ of the collisions. Real gases generally have a rotational relaxation time that varies with temperature. This may be simulated by variations in the fraction Λ but, if detailed balancing is to be satisfied, Λ may have to be a function of the macroscopic temperature, rather than of the energy of the collision pair. This point is investigated in §11.5.

For a gas with coefficient of viscosity proportional to temperature to the power ω, eqn (4.58) can be written in terms of the translational energy E_t in the collision as

$$\bar{Q} \propto \int_0^\infty Q\, E_t^{3/2-\omega} \exp\{-E_t/(kT)\}\, \mathrm{d}E_t\,,$$

so that, by analogy with eqn (3.3), the distribution function for E_t is

$$f_{E_t} \propto E_t^{3/2-\omega} \exp\{-E_t/(kT)\}\,. \tag{5.13}$$

The internal energy E_i of a collision pair is the sum of the internal energies of the collision partners, i.e.

$$E_i = \varepsilon_{i,1} + \varepsilon_{i,2}\,. \tag{5.14}$$

The distribution function for the internal energy of a molecule with ζ internal degrees of freedom may be written (Hinshelwood 1940),

$$f_{\varepsilon_i} \propto \varepsilon_i^{\zeta/2-1} \exp\{-\varepsilon_i/(kT)\}\,. \tag{5.15}$$

We now require the distribution function f_{E_i} for E_i. Consider the fraction of collision pairs with a particular value $\varepsilon_{i,1}$ of internal energy in molecule 1, and therefore with $E_i - \varepsilon_{i,1}$ in molecule 2. Using eqns (5.14) and (5.15), and noting that $\mathrm{d}E_i = \mathrm{d}\varepsilon_{i,2}$ for fixed $\varepsilon_{i,1}$, this fraction is proportional to

$$\varepsilon_{i,1}^{\zeta/2-1} (E_i - \varepsilon_{i,1})^{\zeta/2-1} \exp\{-E_i/(kT)\}\, \mathrm{d}\varepsilon_{i,1}\, \mathrm{d}E_i\,. \tag{5.16}$$

The total fraction of collision pairs with internal energy in the range E_i to $E_i + dE_i$ is proportional to the expression obtained by integrating eqn (5.16) over all $\varepsilon_{i,1}$ from 0 to E_i. Therefore,

$$f_{E_i} \propto E_i^{\zeta-1} \exp\{-E_i/(kT)\}. \tag{5.17}$$

The total energy E_c in the collision is the sum of the relative translational energy and the combined internal energy i.e.

$$E_c = E_t + E_i. \tag{5.18}$$

The probability density of E_t and E_i is proportional to the product of f_{E_t} and f_{E_i} i.e.

$$E_t^{3/2-\omega} E_i^{\zeta-1} \exp\{-(E_t + E_i)/(kT)\}$$

or, using eqn (5.18),

$$E_t^{3/2-\omega} (E_c - E_t)^{\zeta-1} \exp\{-E_c/(kT)\}.$$

The effective temperature T is defined by the total energy E_c in the collision, so that the exponential term may be regarded as a constant. The probability of a particular value of the translational energy is therefore

$$P = C E_t^{3/2-\omega} (E_c - E_t)^{\zeta-1}. \tag{5.19}$$

where C is a constant.

The maximum value of this probability is readily shown to occur for

$$\frac{E_t}{E_c} = \frac{3/2-\omega}{\zeta+\frac{1}{2}-\omega} \tag{5.20}$$

and is

$$P_{max} = C(3/2-\omega)^{3/2-\omega}(\zeta-1)^{\zeta-1}\{(\zeta+\tfrac{1}{2}-\omega)E_c\}^{\zeta+\frac{1}{2}-\omega}.$$

The ratio of the probability to the maximum probability is, therefore,

$$\frac{P}{P_{max}} = \left\{\frac{\zeta+\frac{1}{2}-\omega}{3/2-\omega}\left(\frac{E_t}{E_c}\right)\right\}^{3/2-\omega}\left\{\frac{\zeta+\frac{1}{2}-\omega}{\zeta-1}\left(1-\frac{E_t}{E_c}\right)\right\}^{\zeta-1}. \tag{5.21}$$

A post-collision value of E_t^* is chosen at random from the range 0 to E_c. The ratio P/P_{max} is evaluated by eqn (5.21) and is compared with a random fraction R_f that is generated from a uniform distribution between 0 and 1. This value of E_t^* is employed for the collision if the probability ratio is greater than R_f, but a new value is chosen and the process is repeated if the ratio is less than R_f. This is an application of the *acceptance–rejection method* for sampling from a prescribed distribution, that is discussed in Appendix C.

For a gas with $\omega = \frac{1}{2}$ and two internal degrees of freedom, the equations take on a particularly simple form. The most probable value of E_t is $\frac{1}{2}E_c$ and the probability ratio is

$$P/P_{max} = 4(E_t/E_c)(1-E_t/E_c).$$

Note that the analysis applies equally to the VHS and VSS models and the inverse power law model, including the hard sphere and Maxwell cases.

The total energy in the collision is conserved and the post-collision internal energy and the post-collision value of the pair internal energy is, from eqn (5.18)

$$E_i^* = E_c - E_t^*. \tag{5.22}$$

This must now be divided between the two molecules. The post-collision total internal energy is a constant for this process, and eqn (5.16) shows that the probability that molecule 1 has a post-collision internal energy $\varepsilon_{i,1}^*$ is

$$P = D\,(\varepsilon_{i,1}^*)^{\zeta/2-1}\,(E_i^* - \varepsilon_{i,1}^*)^{\zeta/2-1}, \tag{5.23}$$

where D is a constant.

The maximum probability occurs when the internal energy is equally divided between the two molecules, and the ratio of the probability to the maximum probability is

$$P/P_{max} = 2^{\zeta-2}\,(\varepsilon_{i,1}^*/E_i^*)^{\zeta/2-1}\,(1 - \varepsilon_{i,1}^*/E_i^*)^{\zeta/2-1}. \tag{5.24}$$

The acceptance–rejection method may now be applied in a similar manner to the selection of the post-collision relative translational energy. The special case of two internal degrees of freedom is such that all values of the single molecule post-collision internal energy are equally possible.

The post-collision internal energy may be directly assigned to the two molecules. The post collision relative speed in the centre of mass frame of reference is

$$c_r^* = (2E_t^*/m_r) = 2(E_t^*/m)^{1/2}. \tag{5.25}$$

A new direction for this speed may be chosen at random for the VHS and hard sphere models, from eqn (2.35) for the VSS model, while, for the inverse power law model, eqn (2.22) for the post-collision velocity components is readily modified to allow for the energy change.

While the preceding equations are adequate for the application of the Larsen–Borgnakke model to numerical studies, its application to analytical studies requires explicit expressions for the distribution functions. Eqn (5.19) may written as the probability of E_t/E_c (E_c is regarded as a constant in this distribution), and the constant may then be determined from the condition that the integral of this probability with respect to E_t/E_c over the interval 0 to 1 must be unity. The modified constant is then

$$C\,E_c^{5/2-\omega+\zeta} = 1/B(5/2-\omega,\zeta) = \Gamma(5/2-\omega+\zeta)/\{\Gamma(5/2-\omega)(\Gamma(\zeta)\},$$

and the distribution function for the fraction of translational energy in the collision is (Haas et al 1993)

$$f\left(\frac{E_t}{E_c}\right) = \frac{\Gamma(5/2-\omega+\zeta)}{\Gamma(5/2-\omega)\Gamma(\zeta)}\left(\frac{E_t}{E_c}\right)^{3/2-\omega}\left(1-\frac{E_t}{E_c}\right)^{\zeta-1}. \tag{5.26}$$

The corresponding distribution function for the internal energy fraction is

$$f\left(\frac{E_i}{E_c}\right) = \frac{\Gamma(5/2-\omega+\zeta)}{\Gamma(5/2-\omega)\Gamma(\zeta)}\left(1-\frac{E_i}{E_c}\right)^{3/2-\omega}\left(\frac{E_i}{E_c}\right)^{\zeta-1}. \tag{5.27}$$

A similar procedure can be applied to the probability expressed in eqn (5.23) to give the distribution function for the fraction of the post-collision internal energy in the species 1 molecule as

$$f\left(\frac{\varepsilon^*_{i,1}}{E_i^*}\right) = \frac{\Gamma(\zeta)}{\{\Gamma(\zeta/2)\}^2}\left(\frac{\varepsilon^*_{i,1}}{E_1^*}\right)^{\zeta/2-1}\left(1-\frac{\varepsilon^*_{1,1}}{E_i^*}\right)^{\zeta/2-1}. \tag{5.28}$$

Note that the first distribution function becomes unity for the special case of two internal degrees of freedom. A similar distribution can be applied to any division of the internal energy of a single molecule into the energies of separate internal modes.

The preceding analysis has been essentially for the power-law models that include the VHS model, but Hassan and Hash (1993) have shown that it can be applied also to the GHS model. The distribution function for E_t can be written from eqn (4.59) as

$$f_{E_t} = \left(\frac{2}{\pi}\right)^{1/2}\frac{2}{m_r^2}\left(\frac{m_r}{kT}\right)^{3/2}\frac{1}{\overline{\sigma_T c_r}}\sigma_T E_t \exp\left(-\frac{E_t}{kT}\right). \tag{5.29}$$

The product of f_{E_t} and f_{E_i} is now proportional to

$$\sigma_T E_t (E_c - E_t)^{\zeta-1}\exp\{-E_c/(kT)\}$$

and the probability of a particular value of the translational energy is now

$$P = C\sigma_T E_t (E_c - E_t)^{\zeta-1}, \tag{5.30}$$

where C is a constant (different from that in eqn (5.19)). This equation may be combined with the definition of the GHS model in eqn (3.80), to give

$$P = C'\{\alpha_1(E_t/\varepsilon)^{1-\psi_1} + \alpha_2(E_t/\varepsilon)^{1-\psi_2}\}\{(E_c/\varepsilon)-(E_t/\varepsilon)\}^{\zeta-1}, \tag{5.31}$$

where C' is another constant. The value of E_t at the maximum value of this probability is given by the solution of

$$\begin{aligned}\alpha_1(E_t/\varepsilon)^{-\psi_1}\{(\psi_1-\zeta)(E_t/\varepsilon) + (1-\psi_1)(E_c/\varepsilon)\} \\ + \alpha_2(E_t/\varepsilon)^{-\psi_2}\{(\psi_2-\zeta)(E_t/\varepsilon) + (1-\psi_2)(E_c/\varepsilon)\} = 0\end{aligned} \tag{5.32}$$

and P_{max} is found by substituting this value into eqn (5.31). An iterative solution is therefore required to determine P/P_{max} in order to apply the acceptance–rejection method to the selection of a representative value of E_t. The remaining steps in the Larsen–Borgnakke method for the division of the internal energy between molecules and the division between modes within each molecule are unchanged from those for the power-law models.

5.4 The Larsen–Borgnakke model in a gas mixture

The two molecules in a collision may differ, not only in mass and diameter, but also in the number of internal degrees of freedom and the temperature exponent of the coefficient of viscosity. The latter difference can lead to mathematical difficulties but, if a separate exponent ω_{12} is determined for each combination of molecules in the collision, the simple gas theory is readily extended to the gas mixture. The distribution function for the relative translational energy is changed only by the replacement of ω by ω_{12}. Equation (5.15) applies if ζ is replaced by ζ_1 for the number of internal degrees of freedom of molecule 1. Eqn (5.16) then becomes

$$\varepsilon_{i,1}^{\zeta_1/2-1}(E_i-\varepsilon_{i,1})^{\zeta_2/2-1}\exp\{-E_i/(kT)\}\,\mathrm{d}\varepsilon_{i,1}\,\mathrm{d}E_i\,. \tag{5.33}$$

The integration of this equation over all values of the internal energy of molecule 1 now gives the following distribution function for E_i

$$f_{E_i} \propto E_i^{\bar{\zeta}-1}\exp\{-E_i/(kT)\}\,, \tag{5.34}$$

where $\bar{\zeta}=(\zeta_1+\zeta_2)/2$ is the average number of internal degrees of freedom. The selection of the post-collision value of the translational energy is then unchanged from the simple gas case, as long as ζ is replaced by $\bar{\zeta}$ and ω is replaced by ω_{12}. However, the singularity at $\zeta=1$ must be taken into consideration in the case of a gas mixture. For example, the mean number of degrees of freedom is unity in the collision of a diatomic molecule with a monatomic molecule. Also, the rotational relaxation rate varies with the molecular species, and the collision between a diatomic molecule for which the rotational energy is to change with a similar molecule for which it is not to change can also lead to a value of unity for the mean number of effective degrees of freedom. The maximum probability then occurs when E_t is equal to E_c, and the probability ratio is

$$P/P_{\max}=(E_t/E_c)^{3/2-\omega_{12}}. \tag{5.35}$$

The distribution function for the ratio of the translational energy to the total energy in a collision between unlike molecules is

$$f\left(\frac{E_t}{E_c}\right)=\frac{\Gamma(5/2-\omega_{12}+\bar{\zeta})}{\Gamma(5/2-\omega_{12})\Gamma(\bar{\zeta})}\left(\frac{E_t}{E_c}\right)^{3/2-\omega_{12}}\left(1-\frac{E_t}{E_c}\right)^{\bar{\zeta}-1}. \tag{5.36}$$

The equation simplifies when the mean number of participating internal degrees of freedom is unity.

The equations for the division of the internal energy between the two molecules are more strongly affected. The equation for the probability of internal energy in molecule 1 becomes

$$P=D\,(\varepsilon_{i,1}^{*})^{\zeta_1/2-1}(E_i^{*}-\varepsilon_{i,1}^{*})^{\zeta_2/2-1}.$$

The maximum probability is when the internal energy is

$$\varepsilon_{i,1}^{*}=\tfrac{1}{2}E_i^{*}(\zeta_1/2-1)/(\bar{\zeta}/2-1)$$

and the ratio of the probability to the maximum probability becomes

$$\frac{P}{P_{\max}} = \frac{(\bar{\zeta}-2)^{\bar{\zeta}-2}}{(\zeta_1/2-1)^{\zeta_1/2-1}(\zeta_2/2-1)^{\zeta_2/2-1}}\left(\frac{\varepsilon^*_{i,1}}{E_i^*}\right)^{\zeta_1/2-1}\left(1-\frac{\varepsilon^*_{i,1}}{E_i^*}\right)^{\zeta_2/2-1}. \quad (5.37)$$

If both ζ_1 and ζ_2 are equal to two, the internal energy is uniformly distributed, as for the simple gas case. The probability ratio for $\zeta_1 = 2$ is

$$P/P_{\max} = (1 - \varepsilon^*_{i,1}/E_i^*)^{\zeta_2/2-1}, \quad (5.38)$$

and that for $\zeta_2 = 2$ is

$$P/P_{\max} = (\varepsilon^*_{i,1}/E_i^*)^{\zeta_1/2-1}. \quad (5.39)$$

The singularity when ζ_1 and ζ_2 are unequal but have a mean value of two is more difficult. The probability equation for the energy in molecule 1 is

$$P = D\left(\frac{\varepsilon^*_{i,1}}{E_i^* - \varepsilon_{i,1}}\right)^{\zeta_1/2-1}, \quad (5.40)$$

and this is infinite when $\varepsilon^*_{i,1} = E_i^*$ and $\zeta_2/2 - 1$ is positive, or when $\varepsilon^*_{i,1} = 0$ and $\zeta_2/2-1$ is negative. The acceptance–rejection method can be applied if a cut-off value is applied to the energy before the singularity is reached.

The distribution function for the ratio of post-collision energy in the species 1 molecule is

$$f\left(\frac{\varepsilon^*_{i,1}}{E_i^*}\right) = \frac{\Gamma(\bar{\zeta})}{\Gamma(\zeta_1/2)\Gamma(\zeta_2/2)}\left(\frac{\varepsilon^*_{i,1}}{E_i^*}\right)^{\zeta_1/2-1}\left(1-\frac{\varepsilon^*_{i,1}}{E_i^*}\right)^{\zeta_2/2-1}. \quad (5.41)$$

The preceding theory may be applied separately to each pair of species in a multiple mixture.

5.5 The general Larsen–Borgnakke distribution

A general Larsen–Borgnakke distribution function for the division of energy between the translational and internal modes, between molecules, and between internal modes in each molecule may be defined such that it includes all the distribution functions of the preceding sections as special cases. Consider a collision between a molecule of species 1 and a molecule of species 2. The parameter Ξ is defined as the sum of the *average* degrees of freedom. The translational modes contribute $5/2-\omega_{12}$, so that

$$\Xi = (5/2 - \omega_{12}) + \zeta_{\text{rot},1}/2 + \zeta_{\text{rot},2}/2 + \Sigma\zeta_{\text{v},1}/2 + \Sigma\zeta_{\text{v},2}/2. \quad (5.42)$$

The subscripts 1 and 2 refer to the two molecules, while the subscripts r and v refer to the rotational and vibrational modes respectively. The vibrational modes require special treatment and this is presented in the following section. Let Ξ_a be one or more of the terms in eqn (5.42) and let Ξ_b be the *remaining* terms that are participating in the division of energy.

The energy to be assigned to the first group of modes is E_a and E_b is that to be assigned to the second group. The Larsen–Borgnakke result for the distribution of energy between the two groups is

$$f\left(\frac{E_a}{E_a+E_b}\right)=f\left(\frac{E_b}{E_a+E_b}\right)=\frac{\Gamma(\Xi_a+\Xi_b)}{\Gamma(\Xi_a)\Gamma(\Xi_b)}\left(\frac{E_a}{E_a+E_b}\right)^{\Xi_a-1}\left(\frac{E_b}{E_a+E_b}\right)^{\Xi_b-1} \quad (5.43)$$

The sum E_a+E_b of the energy in the two groups is a constant equal to the pre-collision value of the sum, so that the average energy in the first group is

$$\frac{\overline{E_a}}{E_a+E_b}=\frac{\Gamma(\Xi_a+\Xi_b)}{\Gamma(\Xi_a)\Gamma(\Xi_b)}\int_0^1\left(\frac{E_a}{E_a+E_b}\right)^{\Xi_a}\left(1-\frac{E_a}{E_a+E_b}\right)^{\Xi_b-1}\mathrm{d}\left(\frac{E_a}{E_a+E_b}\right).$$

This may be evaluated through eqns (B4), (B37), and (B38), to give

$$\overline{E_a}=\frac{\Xi_a}{\Xi_a+\Xi_b}(E_a+E_b). \quad (5.44)$$

The average energy ratio is therefore equal to the ratio of the number of degrees of freedom, and this confirms that the Larsen–Borgnakke method leads to the equipartition of energy.

For the special case of the assignment of energy to a single internal mode with two degrees of freedom, $\Xi_a=1$, and eqn (5.43) reduces to

$$f\left(\frac{E_a}{E_a+E_b}\right)=\Xi_b\left(1-\frac{E_a}{E_a+E_b}\right)^{\Xi_b-1}. \quad (5.45)$$

This distribution function is amenable to sampling from the inverse cumulative distribution function, and it is shown in Appendix C that the representative value corresponding to the random fraction R_f is

$$\frac{E_a}{E_a+E_b}=1-R_f^{1/\Xi_b}. \quad (5.46)$$

The probability that the energy E_a in the internal mode exceeds some reference value E_d is obtained by integrating eqn (5.45) with respect to $E_a/(E_a+E_b)$ from $E_d/(E_a+E_b)$ to 1, to give

$$P_{E_a>E_d}=\left(1-\frac{E_d}{E_a+E_b}\right)^{\Xi_b}. \quad (5.47)$$

Note that these distributions for the partitioning of energy do not just sample from the local equilibrium distribution. The temperature of the Maxwellian that appears in the derivation of the Larsen–Borgnakke method is based on an 'effective temperature' of the *particular* collision.

The preceding analysis has implicitly assumed that the parameter Ξ in eqn (5.42) includes all modes that are subject to Larsen–Borgnakke energy redistribution in the collision. It has generally been assumed that the total collision energy E_c must include the energy of all modes that are to be subject to redistribution in the collision. The following analysis shows that this is an unnecessary restriction.

Consider the selection of energy ε_1 into a single internal mode with ζ_1 degrees of freedom. The distribution functions are identical if the cumulative distributions are identical, and the analysis is facilitated if we represent the cumulative distribution by the probability $P_{\varepsilon_1 > E_a}$ that ε_1 exceeds some reference value E_a. The sum of the degrees of freedom of the other internal modes that are active for redistribution in the collision is $\Sigma\zeta$. If the collision energy includes that in all internal degrees of freedom, the distribution function for the energy in the mode being considered is, from eqn (5.43),

$$f\left(\frac{\varepsilon_1}{E_c}\right) = \frac{\Gamma(5/2-\omega_{12}+\Sigma\zeta/2+\zeta_1/2)}{\Gamma(5/2-\omega_{12}+\Sigma\zeta/2)\Gamma(\zeta_1/2)}\left(1-\frac{\varepsilon_1}{E_c}\right)^{3/2-\omega_{12}+\Sigma\zeta/2}\left(\frac{\varepsilon_1}{E_c}\right)^{\zeta_1/2-1} \cdot \quad (5.48)$$

The probability that ε_1 exceeds E_a in this particular collision is, therefore,

$$P = \frac{\Gamma(5/2-\omega_{12}+\Sigma\zeta/2+\zeta_1/2)}{\Gamma(5/2-\omega_{12}+\Sigma\zeta/2)\Gamma(\zeta_1/2)}\int_{E_a/E_c}^{1}\left(1-\frac{\varepsilon_1}{E_c}\right)^{3/2-\omega_{12}+\Sigma\zeta/2}\left(\frac{\varepsilon_1}{E_c}\right)^{\zeta_1/2-1} d\left(\frac{\varepsilon_1}{E_c}\right)$$

and eqns (B40) and (B41) enable this to be written

$$P = I_{1-E_a/E_c}(5/2-\omega_{12}+\Sigma\zeta/2, \zeta_1/2), \quad (5.49)$$

where I is the function defined by eqn (B40)

The required probability is obtained by integrating this probability over all collisions in which E_c exceeds E_a. E_c is the sum of the translational energy with the distribution function of eqn (5.13) and the internal energy with the distribution function of eqn (5.27). The distribution function for E_c is the product of the individual functions, and this may be normalized in the usual way. We then have

$$P_{\varepsilon_1 > E_a} = \frac{1}{\Gamma(5/2-\omega_{12}+\Sigma\zeta/2+\zeta_1/2)}\int_{E_a/kT}^{\infty}\left(\frac{E_c}{kT}\right)^{3/2-\omega_{12}+\Sigma\zeta/2+\zeta_1/2} \times I_{1-E_a/E_c}(5/2-\omega_{12}+\Sigma\zeta/2, \zeta_1/2)\exp\left(-\frac{E_c}{kT}\right) d\left(\frac{E_c}{kT}\right). \quad (5.50)$$

If the redistribution is between the translational energy and the mode 1 internal energy only, the probability is given by eqn (5.50) with $\Sigma\zeta$ set to zero. Therefore, the necessary and sufficient condition for the serial redistribution to be valid is that the expression

$$\frac{1}{\Gamma(a+b)} \int_d^\infty c^{a+b-1} \, I_{1-d/c}(a,b) \exp(-c) \mathrm{d}c \tag{5.51}$$

must be independent of a.

For the common case in which the redistribution is to an internal mode with two degrees of freedom, $b = 1$ and eqn (B43) eliminates the incomplete beta function. The expression then becomes

$$\frac{1}{\Gamma(a+1)} \int_d^\infty c^{\,a} (1-d/c)^a \exp(-c) \mathrm{d}c \,,$$

and, if the integration variable is changed to $c-d$, this becomes

$$\frac{\exp(-d)}{\Gamma(a+1)} \int_0^\infty (c-d)^a \exp\{-(c-d)\} \, \mathrm{d}(c-d).$$

The integral is now the gamma function of $a+1$ and the expression reduces to $\exp(-d)$, so that the condition is satisfied. Numerical integration is required for values of b other than 1. This was performed for a number of values of b, both less than 1 and greater than 1. In all cases, it was found that the expression was unaffected by the value of a.

The Larsen–Borgnakke process is therefore unaffected if there is a succession of redistributions, each of which involves only a single internal mode and the translational mode. Although the 'collision energy' in each of these 'serial' redistributions is less than that in a redistribution that involves all energy modes, this is exactly balanced by the greater fraction of the energy that goes into the single internal mode in the serial case. The parameter Ξ_b that appears in eqn (5.45) for the redistribution to a single internal mode with two degrees of freedom is obtained by excluding from Ξ the effective number of degrees of freedom of this mode. This will be shown to be valuable in the quantum vibration model because it removes the need to specify the effective number of degrees of freedom.

5.6 Vibrational and electronic energy

The reason for treating the vibrational energy modes separately from the rotational modes is that the vibrational levels are widely spaced and vibration can rarely be regarded as being fully excited. Advantage can be taken of the wide spacing to treat each level as a distinct species with specified cross-sections for the transitions between levels that occur as a result of intermolecular collisions. However, if a classical model were to be used, the obvious 'physical' model of a diatomic molecule would be two spheres connected by a spring. This would add a number of additional problems to those discussed in §5.1 for the sphero-cylinder model, and the model would almost certainly be inadequate as well as impractical. There seems to be no alternative to a phenomenological approach.

The Larsen–Borgnakke method can be applied to the vibrational modes through either a classical procedure that assigns a continuously distributed vibrational energy to each molecule, or through a quantum approach that assigns a discrete vibrational level to each molecule.

The classical version regards the degree of excitation and the effective number of vibrational degrees of freedom as variables. If it is assumed that the vibration follows the simple harmonic oscillator model, the specific vibrational energy associated with a mode having a characteristic vibrational temperature Θ_v is

$$e_v = \frac{R\,\Theta_v}{\exp(\Theta_v/T)-1}. \tag{5.52}$$

Then, by analogy with eqn (1.31), the effective number of degrees of freedom at temperature T is

$$\zeta_v = \frac{2\,\Theta_v/T}{\exp(\Theta_v/T)-1}. \tag{5.53}$$

Note that, for small x, $\exp(x) = 1+x$, so that the number of degrees of freedom tends to 2 as the temperature becomes large in comparison with the characteristic temperature. A number of vibrational modes may be associated with a molecule, and each of these will have a distinct characteristic temperature. The characteristic temperatures that are associated with typical diatomic molecules are listed in Appendix A. The total vibrational energy is obtained by summing over all modes.

In order to apply the harmonic oscillator model within the Larsen-Borgnakke method, an effective temperature must be determined from the energies in the collision. As noted earlier, the method involves an implicit temperature based on the total energy in the collision. Equipartition is achieved if the number of effective degrees of freedom is determined for each molecule from the total collision energy associated with that molecule. In a collision between a species p molecule with a species q molecule the effective temperature of the species p molecule is given by

$$T_p = \frac{(E_t + \varepsilon_{i,p})/k}{5/2 - \omega_{pq} + \zeta_{rot,p}/2 + \sum_{j=1}^{j_p} \dfrac{\Theta_{v,p,j}/T_p}{\exp(\Theta_{v,p,j}/T_p)-1}} \tag{5.54}$$

where $\zeta_{rot,p}$ is the number of rotational degrees of freedom and j_p the number of vibrational modes of this species. An iterative method must be used to solve this equation and the similar equation for species q for T_p and T_q, respectively. The number of vibrational degrees of freedom for each molecule is then obtained by substituting these temperatures into eqn (5.53) and summing over all modes.

It is also necessary to divide the internal energy ε_i of the individual molecules into the rotational energy ε_r and vibrational energy ε_v. Note that both the subscript to indicate the species and the asterisk superscript to indicate the post-collision values have been omitted because they each apply to all three energies. Each of ε_r and ε_v is distributed in accordance with the distribution function of eqn (5.15), and eqn (5.16) now shows that the probability of a particular value of ε_r is proportional to

$$(\varepsilon_r)^{\zeta_r/2-1}(\varepsilon_i-\varepsilon_r)^{\zeta_v/2-1} \tag{5.55}$$

and

$$\varepsilon_v = \varepsilon_i - \varepsilon_r. \tag{5.56}$$

These equations are similar to the equations that led to eqn (5.37), and the application of the acceptance–rejection procedure leads to similar problems with singularities when one or other of the exponents is zero or negative. These may be dealt with in a similar manner to that proposed in the discussion of eqn (5.37). Alternatively, the distribution between internal energies could have been avoided through the adoption of the serial mode of application of the Larsen–Borgnakke technique. In any case, this classical treatment of the vibrational modes has been effectively superseded by the quantum model.

The quantum versions of the Larsen–Borgnakke method restrict the values of the vibrational energy of a molecule to those that correspond to the discrete quantum levels. For level i of a particular mode, the harmonic oscillator model gives the vibrational energy of that mode as

$$\varepsilon_i = i k \Theta_v. \tag{5.57}$$

Quantum models were introduced by Haas et al (1993) and by Bergemann and Boyd (1994). The following treatment is based largely on the latter model. The Boltzmann distribution of the energy levels i can be written

$$f_{\varepsilon_v} \propto \exp\left(-\frac{\varepsilon_v}{kT}\right)\delta\left(\varepsilon_v - i k \Theta_v\right). \tag{5.58}$$

The Dirac delta function δ is defined such that

$$\int_a^b f(x)\delta(x-x_0) = \begin{cases} f(x_0) & \text{if } a<x_0<b \\ 0 & \text{otherwise.} \end{cases} \tag{5.59}$$

The integer i ranges from 0 to ∞, and the δ function effectively transforms an integration over a continuous distribution of energy into a summation over the discrete energy levels.

Consider the Larsen–Borgnakke distribution of energy into a particular vibrational mode of molecule 1. Advantage may be taken of the option that

allows the serial consideration of the internal modes, as discussed in the final paragraph of the preceding section. The parameter Ξ_b is then $5/2-\omega_{12}$ and the 'collision energy' E_c is the sum of the pre-collision vibrational energy of molecule 1 and the relative translational energy. The energy in the modes other than that under selection is simply the relative translational energy E_t, and the distribution function for E_t is given by eqn (5.13). The distribution function for a particular combination of ε_v and E_t is given by the product of this distribution and that in eqn (5.58), or

$$(E_c - \varepsilon_v)^{3/2-\omega_{12}}\,\delta(\varepsilon_v - i\,k\Theta_v)\exp\{-E_c/(kT)\}.$$

The energy E_c is a constant in this distribution, and the exponential term may again be regarded as a constant. The maximum probability is for the ground state at $\varepsilon_v = 0$, and the ratio of the probability of a particular level to that of the ground state is

$$\frac{P}{P_{max}} = \left(1 - \frac{\varepsilon_v}{E_c}\right)^{3/2-\omega_{12}} \delta(\varepsilon_v - i^* k\Theta_v). \tag{5.60}$$

The analysis has followed the usual convention of regarding the ground state with $i^*=0$ as being at zero energy. In fact, there is a finite zero-point energy of $\frac{1}{2}k\Theta_v$ in this state, and there is no difficulty associated with the setting of a negative lower limit in the definition of the Dirac delta for the ground state. The probability ratio may be written in a discrete form as

$$\frac{P}{P_{max}} = \left(1 - \frac{i^* k\Theta_v}{E_c}\right)^{3/2-\omega_{12}}. \tag{5.61}$$

The acceptance–rejection method is applied to this probability ratio in order to select the energy in the post-collision level i^* of molecule 1. This selection process is applied to potential levels that are chosen uniformly from the integers between 0 and the maximum level

$$i^*_{max} = \lfloor E_c/(k\Theta_{v,1}) \rfloor, \tag{5.62}$$

where $\lfloor . \rfloor$ denotes truncation.

Note that the partial excitation of the gas as a whole comes entirely from the truncation of the vibrational energy to the discrete quantum levels during the selection process. The above analysis employs the equally spaced energy levels of the harmonic oscillator model. Alternatively, the method may be based on lists of the actual anharmonic vibrational levels for each molecular species. Dissociation of the molecule occurs if the selection process yields the vibrational level that corresponds to the dissociation energy. The resulting chemical rate coefficients are derived and discussed in §6.6.

The Boltzmann distribution for the electronic levels gives the following result for the fraction of molecules in level i

$$\frac{N_i}{N} = \frac{g_i \exp\{-\varepsilon_{\mathrm{el},i}/(kT)\}}{\sum g_i \exp\{-\varepsilon_{\mathrm{el},i}/(kT)\}}. \tag{5.63}$$

Here, $\varepsilon_{\mathrm{el},i}$ is the energy of level i and g_i is the degeneracy (or number of states in the level), and the summation is over all electronic levels. The Dirac delta can again be used to write this discrete distribution as a continuous distribution function for $\varepsilon_{\mathrm{el}}$. This is

$$f_{\varepsilon_{\mathrm{el}}} \propto g_i \exp\left(-\frac{\varepsilon_{\mathrm{el}}}{kT}\right)\delta\left(\varepsilon_{\mathrm{el}} - \varepsilon_{\mathrm{el},i}\right). \tag{5.64}$$

The Larsen–Borgnakke method can then be applied in a similar manner to vibration. However, the effect of the degeneracy is best taken into account in the selection of the states prior to the application of the acceptance–rejection procedure. The selection is between the translational and the electronic mode, and the 'collision energy' E_c is the sum of the energies in these modes. The maximum possible level is determined by the truncation of E_c, and the degeneracies are taken into account when selecting a possible state uniformly from the total number of states in the allowed levels. The acceptance–rejection procedure is then applied to the selected state through the probability ratio

$$\frac{P}{P_{\max}} = \left(1 - \frac{\varepsilon_{\mathrm{el},i}}{E_c}\right)^{3/2-\omega_{12}}. \tag{5.65}$$

5.7 Relaxation rates

Consider a gas in which the temperature T_i of a particular internal mode is different from that of the gas as a whole. Intermolecular collisions cause T_i to *relax* towards the final equilibrium temperature. From eqn (5.44), the average value of the energy that is selected by the Larsen–Borgnakke method into a fully excited internal mode is

$$\bar{\varepsilon} = E_c/(5/2-\omega+\bar{\zeta}), \tag{5.66}$$

where E_c is the total collision energy and $\bar{\zeta}$ is the mean number of internal degrees of freedom that contribute to E_c. While E_c is a constant for the selection process that is described by eqn (5.66), it varies from collision to collision and the mean value of E_c may be approximated by

$$\overline{E_c} = (5/2-\omega_{12}+\bar{\zeta})\,kT_e,$$

where T_e is the *final equilibrium* temperature of the gas. Therefore, if the (non-equilibrium) collision rate is ν and the fraction of inelastic collisions is Λ, the overall rate at which energy is transferred to the internal mode is

$$\mathrm{d}\varepsilon_i/\mathrm{d}t = \nu\Lambda kT_e.$$

The rate at which energy is transferred from this mode is

$$d\varepsilon_i / dt = -\nu\Lambda k T_i,$$

so that the net rate of change of the internal temperature is

$$dT_i / dt = \nu\Lambda (T_e - T_i). \tag{5.67}$$

Both ν and Λ may be functions of temperature, and the integration of this equation must be with respect to $\nu\Lambda t$. This gives

$$T_i = T_e - (T_e - T_{i,0}) \exp(-\nu\Lambda t), \tag{5.68}$$

where $T_{i,0}$ is the value of the internal temperature at zero time.

The *relaxation time* is defined (Chapman and Cowling 1970) as the time in which a disturbance to the equilibrium state decays to $1/e$ of its original value. This time is $1/(\nu\Lambda)$, and the relaxation collision number on the 'collision rate time-scale' is given approximately by

$$Z_i = 1/\Lambda. \tag{5.69}$$

A more precise result for the rotational mode (Haas et al 1994) is

$$Z_i = \frac{3\zeta_{rot}(5-2\omega+\zeta_{rot})}{(3+\zeta_{rot})(5/2-\omega)(5/2-\omega+\zeta_{rot})[1-\{1-\zeta_{rot}\Lambda/(5/2-\omega+\zeta_{rot})\}^2]}. \tag{5.70}$$

This is illustrated in Fig. 5.2 for ζ_{rot} =2. The departure of the product ΛZ_i from the approximate value of unity is relatively small for realistic cases.

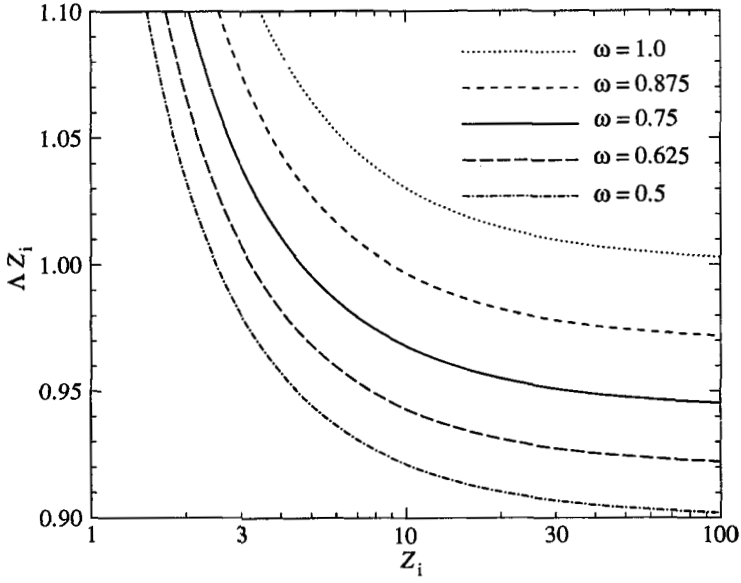


Fig. 5.2 The relationship of the inelastic fraction to the relaxation rate.

5.8 Gas–surface interactions

The majority of problems involve the interaction of gas molecules with solid surfaces. The gas generally has a stream velocity relative to the surface, and this means that the stagnation temperature of the gas differs from its static temperature. In addition, the surface temperature must differ from at least one of these, and the distribution function of the incident molecules will be different from that for the reflected molecules. Therefore, the distribution function of the gas near the surface will not be an equilibrium distribution. Also, the energy of a molecule relative to the surface before it strikes the surface will generally be different from the corresponding energy after it has been reflected from the surface, so that the process is inelastic. The current models for gas–surface interactions are largely phenomenological in nature and their adequacy varies with the nature of the surface and the magnitude of the impact energy of the molecule relative to the surface. The most widely used models are based on generalizations of the diffuse and specular models that were discussed in §4.5 in the context of a gas and surface in equilibrium.

The most common generalization of the diffuse model is to allow the incident and reflected molecules to have different temperatures. For example, it may be assumed that the molecules that are incident on the surface are characteristic of those in an equilibrium gas at temperature T_i, and that those reflected from the surface are characteristic of those in an equilibrium gas at temperature T_r. Note that the temperature T_r may differ from the temperature T_w of the surface. An indication of the extent to which the reflected molecules have their temperature adjusted toward that of the surface is given by the thermal *accommodation coefficient*

$$a_c = (q_i - q_r)/(q_i - q_w). \tag{5.71}$$

Here, q_i and q_r are respectively the incident and reflected energy fluxes, while q_w is the energy flux that would be carried away in diffuse reflection with $T_r = T_w$. The range of a_c is from zero for no accommodation to unity for complete thermal accommodation. Accommodation coefficients may also be defined for the normal and tangential components of momentum. However, the accommodation coefficients may be written as functions of the macroscopic pressure, shear stress, and heat flux and, in general, it is preferable to describe the interaction directly in terms of these quantities.

Experiments with 'engineering' surfaces in contact with gases at normal temperatures indicate that the reflection process approximates diffuse reflection with complete thermal accommodation. This behaviour may be a consequence of such surfaces being microscopically rough with the incident molecules suffering multiple scattering, or of the molecules being momentarily trapped or adsorbed on the surface. Most analytical and numerical studies are based on the assumption of diffuse reflection, and it is fortunate that this appears to be adequate for the vast majority of practical gas flows. The diffuse assumption should be critically reviewed whenever one or more of the following factors are present:

(i) a smooth metal surface that has been outgassed through exposure to high vacuum and high temperatures;

(ii) the ratio of the molecular weight of the gas to that of the surface molecules is small in comparison with unity;

(iii) the translational energy of the molecules relative to the surface is larger than several electron volts.

Measurements of the heat transfer to static gases from heated wires with carefully prepared and thoroughly outgassed surfaces indicate that thermal accommodation coefficients can be significantly less than one. In fact, the thermal accommodation coefficient for light gases on heavy metals surfaces may be very small in comparison with unity (Thomas 1967). Other experiments, using molecular beams directed at similarly prepared surfaces, have shown that the reflected molecules can, on the average, retain some momentum parallel to the surface. These results have been reinforced by measurements in supersonic wind tunnels that employ highly rarefied gases. These indicate that the drag coefficients of bodies such as cones can be significantly lower than the values that would be expected for diffuse reflection. Finally, a measurement of the angular distribution of atomic oxygen from a carbon surface on orbit (Gregory and Peters 1986) found that there was a peak in the reflected number flux in the specular direction. The problem is particularly serious for high-speed flows. As the translational energy of the incident molecules relative to the surface becomes large in comparison with the value corresponding to the surface temperature, complete thermal accommodation is less easily achieved. In addition, for very-high-impact speeds, some of the relative translational energy may be transferred to the internal modes of the incident molecule rather than to the surface. There is also the possibility of chemical reactions, either with surface molecules or with adsorbed gas molecules.

A practical problem that is seriously affected by the above uncertainties is the prediction of the orbital drag of satellites. The translational energy of the atmospheric molecules relative to the surface is approximately 5 eV, compared with a typical 0.025 eV for a molecule of a stationary gas at standard temperature. In addition, the satellite surface is cleaned by outgassing during its long exposure to an ultra high vacuum. The satellite drag coefficient is uncertain and, since this is required in order for the atmospheric density to be inferred from observations of the orbital decay, the atmospheric density is also uncertain.

Any physical model for the interaction of a molecule with a surface would have to be far more complex than the models that have been presented for intermolecular collisions in the gas phase. This applies even to the ideal case of a macroscopically flat surface with no adsorbed gas layer. There is no model of gas surface interactions that is adequate for quantitative studies over a wide range of parameters for all combinations of gases and surfaces, and it is unlikely that one will be forthcoming. If sufficient experimental data were to become available for a particular

application, it could be used to determine the adjustable parameters in an empirical model that satisfies the physical constraints. In the absence of such a model, non-diffuse reflection has most frequently been represented by assuming that some fraction ε of the molecules are reflected specularly, while the remainder are reflected diffusely. However, this model cannot reproduce the molecular beam data that has been obtained for particular cases and there is no justification for the implicit assumption that the result should lie between the limits set by complete diffuse and complete specular reflection. There is, therefore, a need for a more general empirical model.

It is generally accepted that any such model should satisfy the *reciprocity* condition that was defined in eqn (4.81). However, its proof (Wenaas 1971) for the non-equilibrium situation is based on a model of the surface as a phase space array of a large number of identical components. On the assumption that each of these components interacts only once with a gas molecule, the reciprocity condition follows as a consequence of the time reversal invariance of the laws of quantum mechanics. There may be some reservations about the validity of the reciprocity principle when the impact energy of the incident molecule is sufficient to produce a collective reaction among the atoms of the surface lattice. Miller and Subbarao (1971) have verified the reciprocity principle experimentally, but their molecular beam experiments were restricted to a narrow energy range and to cases in which the reflected energy was not very different from the incident energy. Cercignani and Lampis (1971) produced a gas-surface interaction model that satisfies reciprocity. This model has been further developed by Lord (1991a,b), and his results are summarized in the following paragraphs.

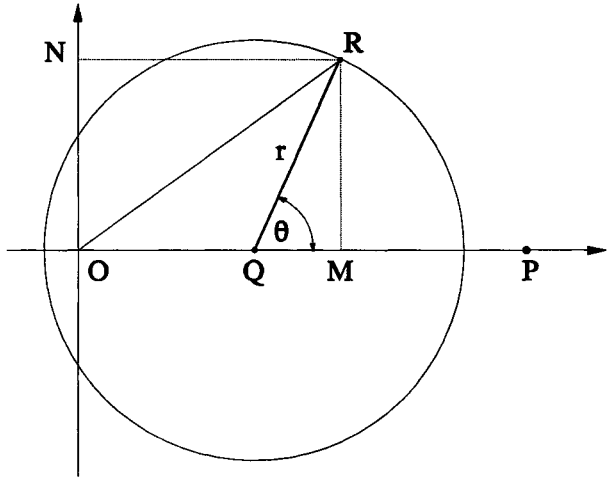


Fig. 5.3 Geometrical representation of the Cercignani–Lampis model.

The model assumes that there is no coupling between the normal and tangential components of the velocity during the reflection process. Set u to be the normal component of the molecular velocity normalized to the most probable molecular speed at the surface temperature, and v and w to be the similarly normalized tangential components.

The probability that an incident molecule with a normalized tangential component v_i is reflected with v_r is

$$P(v_i,v_r) = \{\pi a_t(2-a_t)\}^{-½} \exp[-\{v_r-(1-a_t)v_i)\}^2/\{a_t(2-a_t)\}]. \quad (5.72)$$

Here, a_t is an accommodation coefficient for this component of velocity and $\alpha_t=a_t(2-a_t)$ can be regarded as the accommodation coefficient for the energy of this component normalized to kT. These equations satisfy the reciprocity equation (4.81) and also the normalization condition

$$\int_{-\infty}^{\infty} P(v_i,v_r)\,dv = 1. \quad (5.73)$$

The integral in the normalization condition for the normal component of velocity extends only from 0 to ∞, and the corresponding probability or 'scattering kernel' is

$$P(u_i,u_r) = (2u_r/\alpha_n) I_0\{2(1-\alpha_n)^{½}u_i u_r/\alpha_n\} \exp[\{u_r^2+(1-\alpha_n)u_i^2\}/\alpha_n]. \quad (5.74)$$

It can be shown that the probability function for the magnitude $(v^2+w^2)^{½}$ of the two tangential components also satisfies this equation, as does the function for two components of the angular velocity. Lord introduced the graphical construction of Fig. 5.3 to illustrate this general probability function. The state of the incident molecule may be represented by the point P on the x-axis, and its distance OP from the origin can represent the magnitude of either u_i, $(v_i^2+w_i^2)^{½}$, or ω_i. The point Q represents the average state of the reflected molecules and is situated on OP such that OQ/OP = $(1-\alpha)^{½}$, where the energy accommodation coefficient is either α_n or α_t, as appropriate. The actual state of the reflected molecule is represented by the point R and the probability distribution of this state is given by a two-dimensional Gaussian distribution centred on Q. This shows that the probability of the state lying in the element $r\,dr\,d\theta$ at r,θ is

$$(\pi\alpha)^{-1} \exp(-r^2/\alpha)\,r\,dr\,d\theta, \quad (5.75)$$

where r is the distance QR and θ is the angle PQR. Note that all values of θ are equally probable. The distance OR represents u_r, $(v_r^2+w_r^2)^{½}$, or ω_r, while the projections OM and ON onto the axes represent either v_r and w_r or the the two components of the angular velocity of a reflected diatomic molecule.

A disadvantage of this version of the model is that the accommodation coefficients for the velocity components are directly linked to those for the energies based on these components. This means that it does not include the case of diffuse reflection with incomplete thermal accommodation. Lord (1991) has further extended the model to include this case.

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